

Lecture Notes on Agentic AI

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1 Introduction to Agentic AI

In this first lecture, we introduce the course and build the conceptual foundation for everything that follows. The chapter has the following learning objectives:

1. Understand the evolution from conversational AI to autonomous agents
2. Define what an agent is, independent of any particular technology
3. Explain how Large Language Models (LLMs) became the reasoning core of modern agents
4. Distinguish an agent from a simple chatbot
5. Preview the semester’s roadmap

1.1 From Conversational AI to Autonomous Agents

Over the past few years, our interaction with AI systems has gone through a clear progression. It is worth tracing this progression before we define anything formally, since it explains *why* agentic AI emerged when it did.

1. **Stage 1 – Single-turn question answering.** Early chatbots answered one question at a time, with no memory of what came before and no ability to do anything beyond generating text.
2. **Stage 2 – Multi-turn conversation.** Systems like early ChatGPT could hold a conversation, remembering earlier turns within the same session. Still, the system could only talk – it could not act on the world.
3. **Stage 3 – Tool-augmented assistants.** Models gained the ability to call external tools: a calculator, a search engine, a code interpreter. The model could now affect something outside the conversation, even if only in narrow, developer-defined ways.
4. **Stage 4 – Agentic AI.** The model is given a goal, some degree of autonomy, and the ability to plan multiple steps, use tools, remember context across a task, and sometimes coordinate with other agents, in order to complete that goal with minimal step-by-step human instruction.

This progression is the throughline of the entire course: we are studying what changed, technically, to make Stage 4 possible, and how to build systems that operate at that stage.

1.2 What is an Agent?

Before we talk about *AI* agents specifically, it helps to ask a more general question: what is an agent, independent of any particular technology?

Definition

An **agent** is anything that perceives its environment through sensors and acts upon that environment through actuators, in pursuit of some goal.

This definition says nothing about LLMs, and deliberately so. A thermostat is an agent: it perceives temperature and acts by turning heating on or off. A chess engine is an agent: it perceives the board state and acts by choosing a move. A self-driving car is an agent: it perceives the road through cameras and lidar, and acts through steering, acceleration, and braking.

Example

Is a calculator an agent? No. A calculator has no persistent goal it is pursuing, and it does not perceive an environment – it simply transforms an input you give it into an output, once, with no autonomy over what happens next. Contrast this with a thermostat, which continuously perceives temperature and decides *on its own* when to act, in pursuit of a goal (maintaining a target temperature) that persists over time.

1.3 Where LLMs Fit In

If agents have existed for decades, what changed? The answer is: LLMs gave us a new, general-purpose way to implement the *reasoning and decision-making* component of an agent.

Historically, building an agent meant hand-coding rules (a simple reflex agent) or training a narrow policy for one specific task (a reinforcement learning agent that only knows how to play one game). Both approaches produce agents that are excellent at one thing and useless at almost everything else.

An LLM, by contrast, can reason in natural language across almost any domain it has been trained on. Wrap an LLM with the ability to perceive input, choose actions (including calling tools), and observe the results of those actions, and you have the reasoning core of a general-purpose agent.

Definition

Agentic AI refers to the use of a Large Language Model as the reasoning core of an agent, combined with the ability to perceive context, use tools, retain memory, and take multi-step actions toward a goal with a meaningful degree of autonomy.

1.4 Agent vs. Chatbot: A Concrete Distinction

Students often conflate “using an LLM” with “building an agent.” These are not the same thing, and the distinction matters throughout this course.

Question

Is every LLM application an agent? No. A single-turn chatbot that answers a question and stops is not an agent – there is no perceive-act loop, no autonomy over multiple steps, and no persistent goal being pursued. An application only becomes agentic when the model can take an action, observe the result, and decide what to do next *on its own*, repeating this cycle until the goal is reached or it determines it cannot proceed further.

Table 1 summarizes the key differences.

Property	Chatbot	Agent
Turns	Single response	Multi-step loop
Tools	None (usually)	Can call tools/APIs
Autonomy	None – waits for user	Decides next action itself
Memory	Conversation only	Can persist across a task
Goal	Answer this message	Complete an objective

Table 1: Chatbot vs. Agent

1.5 A Mystery to Keep in Mind

Here is a fact that should feel surprising, and that we are not yet equipped to fully explain:

Example

The Kimi K3 Puzzle Kimi K3 is a language model with 2.8 trillion parameters. A recent open-source project demonstrated that this model can run on a single laptop CPU with only 8 GB of RAM.

At first glance this seems impossible – a model of that size, stored naively, would require well over a terabyte of memory. How can it run on a machine with 8 GB?

We do not yet have the tools to answer this properly. It requires understanding two ideas we have not covered: how a transformer is actually structured (Lecture 7), and a technique called *Mixture-of-Experts*, which we will cover once we have that foundation in place. Keep this question in the back of your mind – we will return to it.

1.6 Course Roadmap

This course is organized into four broad movements, spread across 45 lecture hours:

1. **Foundations** (Lectures 1–15): agent taxonomy, transformer architecture, how LLMs are trained and adapted, and your first hands-on agent built from scratch.
2. **Core techniques** (Lectures 16–28): the LLM training pipeline in depth, retrieval-augmented generation, and reinforcement learning.
3. **Protocols and integration** (Lectures 29–41): the Model Context Protocol (MCP) and Agent-to-Agent (A2A) protocol, reasoning patterns, tool use, planning, memory, multi-agent systems, and modern agent frameworks.
4. **Evaluation and the frontier** (Lectures 42–45): how to evaluate agents rigorously, a light introduction to self-improving systems, and real-world applications.

Throughout the semester, you will incrementally build a single agent, starting from a hand-written reasoning loop in Lecture 12 and ending with a fully integrated, multi-tool agent in Lecture 41.

1.7 Summary

We traced the evolution from single-turn chatbots to agentic AI, defined an agent independently of any specific technology, and explained how LLMs came to serve as the general-purpose reasoning core of modern agents. We distinguished a chatbot from a true agent using the perceive-act loop, and previewed the semester's roadmap. We also left ourselves a puzzle – how a 2.8-trillion-parameter model can run on 8 GB of RAM – that we will resolve once we have covered transformer architecture and Mixture-of-Experts models.

Exercises

1. **Agent or not?** For each of the following, state whether it is an agent according to the definition given in this lecture, and justify your answer: (a) a spam filter, (b) a GPS navigation app that only shows a route when asked, (c) a Roomba robot vacuum, (d) a single-turn translation tool.
2. **Stages of evolution.** Give one real example (an app, product, or system you have used) for each of the four stages described in Section 1.1.
3. **Chatbot vs. agent.** You are using an LLM-powered customer support system. Describe one scenario in which it is behaving as a plain chatbot, and one scenario in which the same system would need to behave as an agent to complete the task.
4. **Conceptual.** Explain, in your own words, why an LLM by itself (with no tools, no loop, no memory beyond one conversation) is not an agent under the definition given in this lecture.
5. **Looking ahead.** Based only on what has been discussed so far, write down your own guess for how a 2.8-trillion-parameter model could run on 8 GB of RAM. We will revisit your guess after Lecture 10.